Sandbox VR: Inventing the future of VR gaming experiences with multi-cluster dynamism

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About Sandbox VR

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Hong Kong-based Sandbox VR offers socially immersive VR experiences that combine full-body motion capture and high-quality haptics for unprecedented realism. Groups of up to six friends can freely roam VR spaces together, exploring exclusive virtual worlds, and relying on each other to succeed in games designed to be social experiences. Sandbox VR now operates in 47 global locations and is expanding rapidly.

Industries: Gaming

Google Cloud results

- Supports business growth by enabling zero platform downtime at peak hours
- Builds customer loyalty through ultra-low latency website video connectivity
- Optimizes gaming performance through automated monitoring of at least 40 virtual reality machines in each location
- Boosts conversion

Autos
norma
peak v
hours

Location: Hong Kong

Storage (https://cloud.google.com/storage/),

(https://cloud.google.com/kubernetes-engine/),

(https://cloud.google.com/pubsub/)

Google Cloud (https://cloud.google.com/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud Storage (https://cloud.google.com/storage/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

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Cloud Pub/Sub (https://cloud.google.com/pubsub/)

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rate by 50 through BigQuery-enabled A/B testing

Endorsed by celebrities and fans around the world, Sandbox VR (https://sandboxvr.com/hongkong/) drives the future of virtual reality (VR) gaming experiences. With groundbreaking full-body motion capture and haptics technology that bring unprecedented realism and social immersion to gaming, Sandbox VR offers a unique experience that blurs fantasy and reality.

Founded by entrepreneur Steve Zhao, the Hong Kong-based startup witnessed remarkable growth since the pandemic. As demand for VR gaming experiences in physical stores returned, the company expanded from 15 venues to 47 across Asia, Europe, and North America. The popularity of games, such as Deadwood Valley (https://sandboxvr.com/hongkong/experience/deadwoodvalley), paved the way for Sandbox VR to create a VR game inspired by the smash hit series Squid Game with Netflix, opening exciting new paths for fan engagement.

"Our expansion journey, while exciting, has also posed challenges to our lean team of developers and engineers. Google Cloud has proven to be the optimal choice for automating operational tasks at scale and driving high-level business intelligence, to continuously reinvent the VR gaming experience."

—Hanks Mak, Senior Director, Software Engineering, Sandbox VR

Amid booming popularity, Sandbox VR wanted a robust cloud infrastructure that can support essential business functions, hosted on an on-premise environment, by a lean team of two DevOps engineers. These included event bookings, video management, and VR machine monitoring. The company also needed a world-class data warehouse to power A/B testing of trailers and other content on its website to boost conversion rates.

Eventually, Sandbox VR chose Google Cloud (https://cloud.google.com/) thanks to its comprehensive suite of integrated solutions and user-friendly interface that drives productivity. The VR firm turned to Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE) for a dynamic microservices environment, and BigQuery (https://cloud.google.com/bigquery) for driving machine learning insight into user behavior.

"Our expansion journey, while exciting, has also posed challenges to our lean team of developers and engineers," says Hanks Mak, Senior Director, Software Engineering, Sandbox VR. "Google Cloud has proven to be the optimal choice for automating operational tasks at scale and driving high-level business intelligence, to continuously reinvent the VR gaming experience."

Powering unlimited global expansion with a leading-edge microservices cloud architecture

With an increasing network of stores globally, Sandbox VR sought to build a multi-cluster microservices architecture to enable the highest availability regardless of location. It deployed Multi Cluster Ingress (https://cloud.google.com/kubernetes-engine/docs/concepts/multi-cluster-ingress) for GKE, a cloud hosted controller for GKE clusters. This allows API requests to be distributed across Kubernetes clusters in different regions via shared load balancing, thus ensuring optimal availability and scalability.

The firm also relies on GKE for internal platform monitoring, allowing it to oversee store performance, aggregate application logs, and gather metrics. GKE enables Sandbox VR to automate all of the monitoring of at least 40 machines in every store, and manage its performance logs even with limited engineering resources. This ensures that every venue operates optimally.

In addition, Sandbox VR deploys Pub/Sub (https://cloud.google.com/pubsub/docs/overview) as an asynchronous messaging service to facilitate smooth communication between GKE microservices, while boosting reliability of datastreams. This is crucial for maintaining the agility and resilience of the Sandbox VR platform. Sandbox VR also relies on Cloud Storage (https://cloud.google.com/storage) to host unlimited numbers of videos created from Sandbox VR venues around the world from fan reactions to game sequences.

"Through the integrated cloud ecosystem of Google Cloud, which includes GKE, Pub/Sub and Cloud

Storage, we were able to automate business processes, freeing the team to focus on creative platform building," says Marco Lam, Senior DevOps Engineer, Sandbox VR.

Gaming enthusiasts using Sandbox VR gear

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VR gaming experiences with zero downtime anywhere in the world

By implementing GKE Multi Cluster Ingress, Sandbox VR has enabled seamless operations across global clusters with zero downtime. Cloud CDN (https://cloud.google.com/cdn) also enhances the user experience by ensuring the lowest latencies for website videos, such as game trailers, for visitors to watch anywhere in the world, thanks to the global edge network of Google Cloud.

Crucially, GKE autoscaling has permitted Sandbox VR to efficiently handle fluctuating demand on IT resources, ensuring optimal performance during peak hours, while scaling down just as nimbly when usage falls. That means the infrastructure can scale to support three to four times its regular traffic during weekends without human intervention.

"Due to the popularity of our VR gaming experiences, we can see dramatic spikes in demand on our website, especially over weekends when people are the most eager for entertainment," says Mak. "With Google Cloud, we have complete peace of mind that we can easily absorb any surge in booking demand, anywhere in the world, thanks to the nimble autoscaling of GKE."

Sandbox VR gaming area

Unlocking powerful audience insight with BigQuery

One of the biggest benefits of the Google Cloud adoption is the deployment of BigQuery and Google Analytics 4
(https://developers.google.com/analytics/devguides/collection/ga4)
. This has helped Sandbox VR gain a deeper understanding of customer behavior to boost user engagement, conversion rates, and overall platform performance.

By carrying out the extract, transform, and load (ETL) operations to combine user data from multiple global sources, and feed it into BigQuery, Sandbox VR can obtain high-level audience insight to make informed business decisions. In particular, Google Analytics 4 enables Sandbox VR to measure traffic and engagement across its website. Thanks to its native integration with BigQuery, unlimited streams of Google Analytics 4 user traffic data can be run through BigQuery machine learning models for driving business intelligence.

"We're continuously striving to attract new fans into our VR gaming world through our evolving website. BigQuery and Google Analytics 4 are powerful tools that give us precise knowledge of what appeals to our audience, so we can create a more focused offering to grow our customer base."

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One use case is the execution of A/B testing of content, such as website trailers, which allows the firm to monitor conversions to booking and optimize the customer conversion process. Deploying BigQuery and Google Analytics 4 for advanced data analytics has seen Sandbox VR increasing its global conversion rate by 50%.

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Driving an AI-enabled future for immersive VR gaming

Sandbox VR is planning to deploy artificial intelligence (AI) solutions to improve and personalize user journeys, while enhancing the user experience at physical venues. One key application will be the highlight videos the company offers to each visitor after their VR gaming experience.

Currently, this video is created from a template of game sequences. In the future, Sandbox VR envisages leveraging AI to detect cues for when players are most engaged, such as frenetic movement or increased screaming, to automate a more personalized highlights reel. Mak says he envisions Google Cloud tools, such as Vertex AI (https://cloud.google.com/vertex-ai) and Vision AI (https://cloud.google.com/vision), to play a significant role in this evolution.

"As we reinvent immersive gaming, we anticipate partnering more with Google Cloud on cutting-edge tools, including advanced AI solutions, to help us provide gamers worldwide with unforgettable experiences. There's no limit to the worlds we can create to make the future of VR gaming even more socially immersive."

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Sandbox VR is also experimenting with BigQuery Studio (https://cloud.google.com/blog/products/data-analytics/announcing-bigquery-studio), a new collaborative analytics workspace to accelerate data-to-AI workflows, to empower users to discover and predict data trends. This will continuously improve gaming experiences and provide more incisive customer insights.

"As we reinvent immersive gaming, we anticipate partnering more with Google Cloud on cutting-edge tools, including advanced AI solutions, to help us provide gamers worldwide with unforgettable experiences," says Mak. "There's no limit to the worlds we can create to make the future of VR gaming even more socially immersive and engaging."